

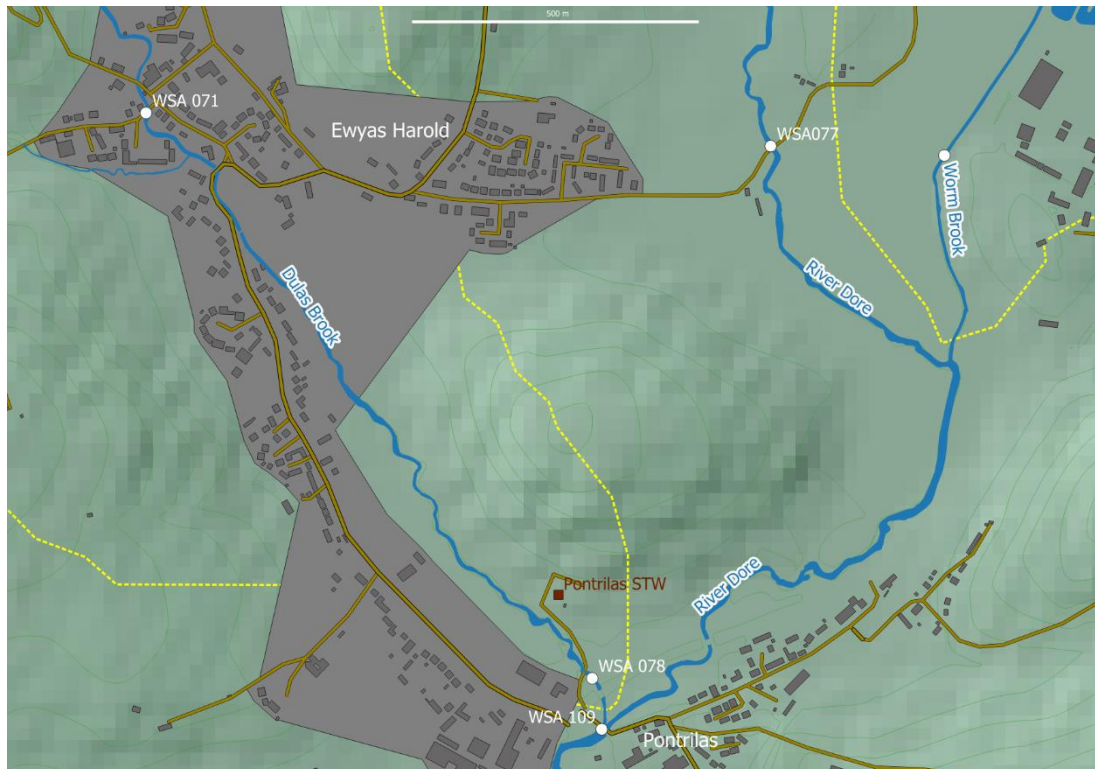


Hot-spot report: Dulas Brook at Pontrilas

A report prepared on behalf of Wye Salmon Association (WSA)

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Author	Gordon Green	

Location



Map showing the Dulas Brook passing through Ewyas Harold en route to the Dore confluence at Pontrilas. White circles are WSA sampling sites.

The Dulas Brook flows through farmland, almost exclusively pasture, until it reaches Ewyas Harold. From there, it flows down to the Dore confluence at Pontrilas. The surrounding area is served by Pontrilas STW (operated by DC-WW; input load 1498 PE), discharging into the Dulas approximately 200m above the Dore confluence.

Hypothesis

That Pontrilas STW exports unacceptable levels of phosphorus and other pollutants into the Dulas Brook.

Test

WSA citizen scientists monitor reactive phosphorus (RP) and electrical conductivity (EC) at two sites on the Dulas, Ewyas Harold (WSA 071) & Pontrilas (WSA 078). These are located upstream and downstream of the STW and will be compared. Paired samples from these sites have been measured within 30 minutes of each other, approximately weekly, over the period Jan-23 to Aug-23.

Results

The results from 22 paired samples are summarised in the table below:

Parameter	Upstream (WSA 071)	Downstream WSA 078)	t-test
Mean RP (mgP/l)	0.04	0.22	0.00%
Mean EC @25°C (µS/cm)	410	449	0.00%

Mean values for reactive phosphorus and electrical conductivity on the Dulas Brook at locations upstream and downstream of Pontrilas STW. The 2-sided, paired t-test gives the probability that the two mean values belong to the same population (i.e., no change).

This tells us that there is a huge (x5) increase in mean RP between the two sites, which is statistically significant with over 99.9% confidence. Mean EC increases by 9%, also with over 99.9% confidence.

Conclusion

- The 5x increase in Dulas RP concentration resulting from the discharge from Pontrilas STW is large and has extremely high statistical confidence. Upstream of the STW, the Dulas RP concentration meets the requirements for WFD *good* classification (<0.072 mgP/l) over the period monitored. Below the STW, the RP concentration corresponds to a *poor* classification only (not even achieving *moderate*). Thus, the STW is causing non-compliance in this section of river, which is a cause for concern. We therefore welcome the fact that Pontrilas STW has been selected to have P-stripping fitted by 2025, within the AMP7 investment cycle.
- The increase in electrical conductivity is equally alarming. Given the fact that most of the EC signal results from natural mineral ions, a 9% increase represents a very significant pollution load. We have separately estimated that the discharge from Pontrilas STW represents approximately 13% of Q95 flow. This is a huge volume of effluent in such a small stream.

Notes on AMP7 improvements:

1. AMP7 will also place tighter limits on BOD, ammonia, and suspended solids – which is to be welcomed.
2. The proposed total phosphorus limit of 1.9 mgP/l represents an approximately 4x reduction from the status quo. However, a rough calculation on the results above suggests that this might not be a sufficient reduction to achieve WFD *good* classification for RP below this STW.